Needs Assessment Questionnaire

StageOne: Indie Musician Discovery Platform

WEB 268 Capstone Project

Prepared By: Eva Q. Cihak — Lead Developer, StageOne
Prepared For: Eric Yoxheimer — Client / Faculty Sponsor
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Developer Contact Information

Eva Quinn Cihak
eqc34019@hawkmail.hacc.edu
(717) 654–5756

Client Contact Information

Eric Yoxheimer
ecyoxhei@hacc.edu
(717) 780–1128

Introduction

The purpose of this Needs Assessment Questionnaire is to define the initial scope and objectives of StageOne, an indie-musician discovery web application, from a business perspective. Based on these business needs, the development team will scope appropriate web-based solutions, balancing cost, schedule, and feature depth. Since there are real tradeoffs between speed of delivery and the quality of features, this document exists to make those tradeoffs explicit before implementation begins.

This questionnaire is organized into three parts:

- Background and Goals
- Audience, Content, and Functionality
- The Field Trip

Part 1: Background and Goals

Company

StageOne is a (fictitious) startup building a web platform that helps independent musicians showcase their work, grow an audience, and connect with local fans. The music-discovery industry is mature but fragmented: streaming giants dominate listening, while smaller platforms serve uploads, fan communities, or direct-to-fan sales separately. StageOne targets the intersection—unsigned or early-career artists who want a single, attractive portfolio site that combines streaming previews, video, gig listings, and fan interaction.

Competitors and brief critique of their sites:

- **Bandcamp** — strong direct-to-fan sales and audio streaming, but a dated, text-heavy UI and limited social/community features.
- **SoundCloud** — excellent audio discovery and comment-on-waveform interactions; weak on professional artist pages and visual branding.
- **ReverbNation** — industry-focused tools (opportunities, charts) but a cluttered UI and heavy paywalls.
- **Spotify for Artists** — strong analytics but only available once an artist is distributed to Spotify; no fan-facing artist home page.
- **BandLab** — collaboration-first with a modern UI; discovery is secondary to creation.

StageOne's opportunity is a clean, modern, AI-assisted discovery experience that puts the artist page front and center.

Who are the decision makers?

Eric Yoxheimer, acting as client and faculty sponsor, is the final decision-maker on scope, acceptance criteria, and grading. Eva Cihak, as lead developer, makes day-to-day implementation decisions within the agreed scope.

Who are the other members of the project team?

StageOne is a solo endeavor. Eva Cihak is the only human developer on the project. An AI assistant (Claude) is used as a collaborator for planning, code review, content drafting, and QA, consistent with the instructor's encouragement to incorporate AI into projects.

What are the roles of everyone involved on the project team?

- **Eva Cihak** — project manager, systems analyst, UX/UI designer, front-end developer (React), back-end developer (Node/Express), database engineer (MongoDB), multimedia producer (images/audio/video), QA tester, and presenter.
- **AI assistant (Claude)** — pair programmer, research aid, drafting support, and QA helper; *<u>does not replace any required developer work</u>*.

- **Eric Yoxheimer** — client, faculty sponsor, and final acceptance authority; reviews status reports and the final deliverable.

What human resources do you have for various stages of the process?

- **Discovery & requirements:** Eva Cihak, with review from Eric Yoxheimer.
- **Design (wireframes, brand, multimedia):** Eva Cihak.
- **Database design (ERD, schema):** Eva Cihak.
- **Implementation (front-end, back-end, AI):** Eva Cihak, supported by the AI assistant.
- **QA and presentation:** Eva Cihak.

Project Deliverables

What is the mission statement or summary of your project?

StageOne's mission is to give independent musicians a polished, modern home on the web — one that combines streaming, video, and community in a single AI-assisted experience that helps fans discover artists who match their taste.

What are the basic goals of this project?

- Build a capstone-worthy, portfolio-grade MERN web application.
- Demonstrate full-stack proficiency: authentication, CRUD across four MongoDB collections, media upload/playback, and responsive UI.
- Integrate at least one meaningful AI feature (recommendation chatbot and/or natural-language search) to meet the instructor's bonus criteria.
- Produce original multimedia assets (images, audio, video) and five or more custom-designed components, demonstrating skills from the WEB design courses.
- Deliver all standard project artifacts: proposal, ERD, storyboard, data-flow diagrams, bi-weekly status reports, and final presentation.

What outcome will make this project successful? How will you measure success?

- Fully functional CRUD on at least three user-facing collections (Artists, Tracks, Gigs).
- Working audio and video playback on artist profile pages.
- AI assistant demonstrably returning useful artist/track recommendations during the final presentation.
- Responsive, accessible UI that passes a design review against WCAG 2.1 AA contrast and keyboard-navigation checks.
- Instructor approval of the final presentation and a final grade at or above the “B+” threshold.

What are your schedule requirements?

The project runs for one academic semester (approximately 15 weeks), with bi-weekly status reports and a final presentation in the last week of class. Tentative milestones:

Weeks 1-2	team formation, proposal, needs assessment
Weeks 3-4	ERD, schema, wireframes, storyboard
Weeks 5-8	backend scaffolding, authentication, core CRUD endpoints
Weeks 9-11	frontend components, multimedia production, AI integration
Weeks 12-13	polish, accessibility pass, content population
Weeks 14-15	final testing, documentation, presentation prep

What is the budget for this project? Is there an acceptable budget range?

The cash budget is $0. The project will rely on free tiers of cloud services (MongoDB Atlas free cluster, Vercel and Render free hosting, Cloudinary free-tier media storage, OpenAI developer credits or a local LLM such as Ollama + Llama 3 as a fallback). An acceptable range is $0–$25 total, covering optional domain registration. The primary investment is time.

Describe any work that has been done toward designing/redesigning a new site.

No prior work exists; this is a greenfield project. Early explorations to date include a moodboard of competitor and benchmark sites (see Part 3), a shortlist of custom components, and a preliminary ERD sketch.

Will the site reinforce an existing branding or marketing strategy? How?

StageOne is a new brand with no existing assets. The site itself is the primary branding vehicle, so branding decisions are made inside this project rather than inherited from a parent organization.

Discuss any identity/branding assets (logos, other artwork, and fonts) or issues.

All brand assets will be created from scratch by Eva Cihak as part of the multimedia requirement. Planned assets:

- A custom wordmark logo evoking stage lights.
- A color palette of deep navy with a warm amber accent, suggesting a stage at night.
- A display sans-serif for titles and a neutral sans-serif for UI.
- Original photography for hero imagery, with appropriately licensed stock as a backup.
- Short audio stings for UI feedback and an intro video trailer for the landing page.

Business Priorities

Ranked in order of importance for StageOne, with brief rationale:

1. **Quality execution (graphics, writing, navigation).** Primary; this is a portfolio piece and polish matters more than anything else.

2. **Creating a community of dedicated visitors.** Discovery and community are the product's reason to exist.
3. **Ease of maintenance.** A solo developer must keep the codebase understandable and low-friction.
4. **Time to market.** Hard semester deadline; non-negotiable.
5. **Sending the message that we know the web and use it appropriately.** The app is itself a résumé and must look current.
6. **A web strategy that fits our marketing strategy.** For a fictitious startup, the site is the marketing strategy.
7. **Membership sign-ups and/or newsletter subscriptions.** Core growth metric for the product concept.
8. **Doing better than our competition on the web.** Competitive positioning via modern UI and AI-assisted discovery.
9. **High traffic in terms of return visitors and page views.** Important conceptually; not meaningfully measurable during a semester.
10. **A web strategy that fits our corporate strategy.** Not applicable beyond the fictitious startup premise.
11. **Staying within the budget.** Trivially met at $0 via free tiers.
12. **Repurposing existing content.** Not applicable; all content is created new.

Part 2: Audience, Content, and Functionality

Audience

What types of visitors do you want to attract?

- **Independent musicians (primary)** — unsigned artists, small bands, and bedroom producers looking for a modern portfolio home.
- **Music fans and listeners (primary)** — casual and enthusiast listeners who enjoy discovering new indie artists.
- **Venue bookers and promoters (secondary)** — local venue staff looking for artists to book for upcoming shows.
- **Capstone evaluators (tertiary, but critical)** — the instructor and class, who must see that the app meets WEB 268 requirements during the final presentation.

What are your goals for each type of visitor?

- **Musicians:** sign up, build a profile, upload tracks and videos, post gigs, and grow a following.
- **Fans:** browse and search, play tracks, follow artists, leave comments, and use the AI recommender to find new music.
- **Venue bookers:** filter artists by genre and city, view artist press kits, and contact artists.

- **Evaluators:** observe full CRUD on three or more collections, multimedia playback, five or more custom components, and a working AI feature.

What are your goals for these products/services?

Increase artist sign-ups, drive follow and engagement actions, and demonstrate that AI-assisted discovery produces high-quality recommendations measured by follow-through from fans.

Content

Where will content come from? Will it be new, repurposed, or both?

Primarily new, user-generated content: artists upload their own tracks, videos, bios, and gig postings. A small amount of curated seed content will be created by Eva Cihak to populate the site during demos — three to five fictional "featured" artists with original or appropriately licensed audio, images, and video clips produced as part of the multimedia deliverable.

How often will you add new content?

Continuously. In production, the platform is updated daily by its users. For the capstone, seed content is added once at launch, with a "What's New This Week" editorial block refreshed weekly during development to demonstrate ongoing activity.

Who will update the content?

- **Artist-owned (tracks, videos, bios, gigs):** artists themselves, via authenticated CRUD forms.
- **Fan-owned (follows, comments):** fans themselves.
- **Editorial / system content (featured banner, help pages):** admin role, operated by Eva Cihak.
- **AI-generated content (recommendations, natural-language responses):** generated on demand by the AI service.

Functionality

What functional requirements do you believe to be necessary?

- **Authentication and accounts:** email/password sign-up and login, role separation for Artist, Fan, and Admin, and JWT session management.
- **Artist profiles:** public, media-rich pages with bios, tags, social links, and a following count.
- **Track upload and playback:** audio upload, cloud storage, and an in-page streaming player with per-track play counts.
- **Video embedding or upload:** either embeds (YouTube, Vimeo) or direct upload of short clips.
- **Gig calendar:** artists post upcoming gigs (date, venue, city, ticket link); filterable.
- **Follow system:** fans follow artists for a personalized feed of new tracks and upcoming gigs.
- **Search:** keyword search across artists, tracks, and gigs, with filters for genre and location.

- **AI recommender/assistant:** a chatbot that answers prompts such as "recommend artists similar to X" or "find me acoustic indie in Harrisburg this weekend," plus natural-language search over the catalog; built on the OpenAI API with retrieval over the MongoDB catalog, with a local LLM fallback.
- **Comments and reactions:** fans comment on tracks and gig posts.
- **Admin moderation:** admin removes inappropriate content and suspends users.
- **Responsive design:** usable on phone, tablet, and desktop.

Who will update these functions?

The developer Eva Cihak maintains the application code. Artists and fans drive most content updates through the UI. Admin moderation is performed by Eva Cihak acting as the administrator during the capstone.

Are there extraordinary security concerns?

- User-uploaded media must be type- and size-checked and stored outside the web root.
- Passwords stored as bcrypt hashes; JWT secrets held in environment variables.
- Rate limiting on authentication and AI endpoints to prevent abuse.
- No real payment data handled in the capstone build; any e-commerce flows are mocked.
- Moderation tooling is required to handle inappropriate uploaded audio or video.

Are there other technical issues or limitations?

- Cold-start latency on free-tier hosting during demos.
  - Mitigated by a pre-warm ping before presentation (roughly a minute prior).
- AI API cost ceiling.
  - AI calls will be cached and rate-limited; a local LLM (Ollama + Llama 3) is a fallback.
- Media upload caps (audio ≤ 20 MB, video ≤ 100 MB) to stay within free-tier limits.

Have you budgeted for hosting and maintenance of the site? If so, what is your budget?

Operational budget during the semester is $0, covered by free tiers of MongoDB Atlas, Render (or Vercel + a separate Node API host), Cloudinary (media), and OpenAI developer credits. Estimated post-semester cost to keep the demo live is under $10 per month.

Who will maintain the site contents?

Eva Cihak maintains application code, configuration, and seed content. Artist and fan content is self-service via the UI.

How will the site be served/hosted?

React front-end hosted on Vercel (static), Node/Express API on Render (web service), MongoDB Atlas (database), and Cloudinary (media). HTTPS everywhere, with CI/CD via GitHub Actions on push.

What types of legacy systems/databases are in place?

None. StageOne is a greenfield build.

What is your long-term plan for the site?

After the semester, the site may be maintained as a live portfolio exhibit. Interest in v2 could jumpstart the addition of direct fan-to-artist tipping and real analytics, but those are way out of scope for the capstone.

Part 3: The Field Trip

This section catalogs reference sites used as benchmarks, inspiration, or direct competitors. Each is examined against the categories from the original questionnaire.

Branding in a similar situation

- **Bandcamp** — established indie-music brand; informs StageOne's tone (artist-friendly, anti-gatekeeper).
- **Tiny Desk, NPR Music** — editorial-indie positioning; informs how featured-artist sections should read.
- **BandLab** — newer brand in the space; shows how to modernize indie-music branding.

Appeal to the same target group of customers

- **SoundCloud** — indie-music fans and creators; comment-on-waveform is a behavior worth borrowing.
- **Audiotree Live** — fans who want discovery of live performances; informs the video-forward artist profile.
- **Bandsintown** — same “fans plus local gigs” audience; informs the gig-calendar UX.

Whether or not you would build the site if you were in a different industry

- **Linktree, Beacons** — the “everything in one place” profile pattern, applicable outside music.
- **Are.na** — minimalist, content-first layout; informs a clean profile aesthetic.
- **Spotify artist pages** — gold standard for single-page artist presentation.

Colors, look-and-feel, user interface, layout

- **Spotify** — dark theme with vivid accent color; direct influence on StageOne's color palette.
- **Apple Music** — large imagery, generous whitespace, editorial feel; informs the featured-artist hero.
- **Bandcamp** — text-heavy and utilitarian; used as a counter-example of what to avoid.

Size of site and size of project

- **Bandsintown** — comparable surface area to StageOne v1 (artists, gigs, follows) without sales; realistic target.
- **ReverbNation** — what happens when scope grows too wide; cautionary tale on feature creep.
- **Linktree** — much smaller surface area; shows the minimum viable artist page.

Publishing model (frequency, novelty of content)

- **Pitchfork** — regular editorial cadence; a weekly “What’s New This Week” block.
- **NPR Music** — themed features; model for curated playlists.
- **Bandcamp Daily** — editorial layered on a marketplace.

Attracting new people to the site

- **Spotify Wrapped** — shareable, newsworthy moments drive traffic; StageOne can mimic with “Top Tracks of the Month” share cards.
- **SoundCloud repost** — built-in virality; informs the follow/repost interactions.
- **Tiny Desk** — novelty/series format as a draw; informs a potential featured-session series.

Competitors' sites

- **Bandcamp** — strong commerce, weak modern UX.
- **SoundCloud** — strong discovery audio, weak artist-page branding.
- **ReverbNation** — strong industry tools, weak UI and paywalls.
- **BandLab** — strong modern UX, weak discovery product.

Quality of content

Pitchfork, NPR Music, and Audiotree set the bar for written and multimedia editorial quality and are the reference points for StageOne's featured-artist posts.

Quality of graphics

Apple Music, Spotify, and Tiny Desk set the bar for hero imagery, typography, and motion.

Interactive tools

- **SoundCloud waveform comments** — interaction to emulate for tracks.
- **Bandsintown “Track Artist” button** — interaction to emulate for following.
- **Spotify “Made For You” carousels** — interaction to emulate for AI recommendations.

General functionality

- **Bandsintown** — location-aware gig discovery.
- **Spotify** — recommendation carousels and personal feed.
- **Bandcamp** — clean embeddable audio players.

Community, special features, responsiveness

- **SoundCloud** — most mature fan–artist dialogue (timed comments).
- **Discord music servers** — most alive fan communities; out-of-scope but informative.
- **Patreon** — direct fan-to-artist support model; future-consideration reference.

Overall favorite sites

- **Spotify artist pages** — for the single-page, media-rich artist snapshot.
- **Bandcamp** — for spirit and artist-friendly ethos.
- **Tiny Desk** — for the belief that a platform can elevate indie artists through curation and production quality.

Sign-off

By signing below, both parties acknowledge that the scope, goals, audience, content, and functionality described in this document reflect a shared understanding of the StageOne project at this point in time. Material changes to scope will be documented in a revision of this needs assessment.

Developer: Eva Quinn Cihak

Signature: ______________________________ Date: ____________

Client: Eric Yoxheimer

Signature: ______________________________ Date: ____________